Few-Shot Heterogeneous Hyperspectral Band Selection via Uncertainty-Aware Meta-Learning

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***Abstract*—Existing cross-scene band selection methods of hyperspectral images (HSIs) are often constrained by the difficulty in balancing sample efficiency with performance. Specifically, fully supervised methods demand extensive labeled data, while zero-shot approaches often suffer from poor performance. Furthermore, these methods are typically restricted to handling homogeneous data. In real-world scenarios, HSI applications are frequently confronted with the scarcity of labeled data and significant data heterogeneity. This heterogeneity is manifested as inconsistencies in the number of bands and spectral ranges across different sensors, which hinder the generalization of existing algorithms. To address these challenges, we propose a few-shot heterogeneous hyperspectral band selection method, termed FHBS-UML, which is the first attempt to tackle few-shot band selection for heterogeneous hyperspectral data. To facilitate robust band selection under sample scarcity, the task is modeled as a sequential decision-making process within a meta-reinforcement learning (meta-RL) framework. To tackle data heterogeneity, a heterogeneous HSI aligner (HSA) is designed to unify structural and feature distributions by employing a local–nonlocal transformer encoder and feature alignment learning. Furthermore, to mitigate unstable intraclass variations in heterogeneous data, we propose a stability-enhanced heterogeneous HSI evaluator, which utilizes an adaptive variance-sensitive prototype to calibrate category confidence for robust evaluation. Finally, the band selection policy is optimized via an uncertainty-aware multiobjective meta-learning strategy, enabling efficient adaptation to few-shot target tasks. Experimental results demonstrate the effectiveness of FHBS-UML, with overall accuracy improvements of 12.0%, 6.7%, and 7.4% on Indian Pines (IP), Salinas (SA), and Pavia University (PU) datasets, respectively, in five-shot cross-scene tasks.**

***Index Terms*—Band selection, few-shot learning, hyperspectral images (HSI), meta-learning.**

Received 28 February 2026; revised 12 June 2026; accepted 22 June 2026. Date of publication 30 June 2026; date of current version 27 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant U2570205, Grant 62271374, Grant 62576260, Grant 62206209, and Grant 62571384; in part by the Natural Science Basic Research Program of Shaanxi under Grant 2025JC-JCQN-090; in part by the Key Research and Development Program of Ordos City (2025) under Grant YF20250288; and in part by the Fundamental Research Funds for the Central Universities under Grant QTZX26130. *(Corresponding author: Di Li.)*

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Digital Object Identifier 10.1109/TGRS.2026.3708772

I. Introduction

HYPERSPECTRAL images (HSI), distinguished by its capability to simultaneously capture continuous spectral signatures and spatial texture features, enables fine-grained material identification. Consequently, it holds significant value in diverse fields, including precision agriculture, environmental monitoring, geological exploration, and military reconnaissance. However, the increasing spectral resolution of hyperspectral sensors results in data characterized by high dimensionality, heavy redundancy, and strong interband correlations. This high dimensionality not only imposes substantial burdens on storage and computation but also induces the "Hughes phenomenon" [1], where classification accuracy degrades with increasing dimensionality under limited training samples. To mitigate these challenges, band selection is widely adopted as a key dimensionality reduction technique to identify the most representative and low-redundancy band subset from the original full-band set [2], [3], [4]. In contrast to feature extraction methods, band selection preserves the physical spectral interpretation of the original data. Therefore, it serves as a critical step in the HSI processing pipeline.

Despite the notable progress of existing algorithms, traditional band selection methods are predominantly tailored for single-scene settings [5]. In such scenarios, training and testing sets are derived from the same image, operating under the assumption that they share consistent data distributions and spectral characteristics. This paradigm typically relies on extensive manual annotations specific to the target scene to capture its unique features, making the trained models difficult to directly generalize to unseen scenes where spectral distributions inevitably shift. Such dependency significantly restricts the flexibility and scalability of these algorithms in real-world applications. To overcome this limitation and alleviate the reliance on target data annotation, cross-scene band selection methods have recently emerged [6]. Grounded in transfer learning [7] or domain adaptation theories [8], these approaches exploit shared spectral patterns or structural features across different scenes. By leveraging prior knowledge from a source domain to guide the selection process in the target domain, they facilitate robust feature selection even when target samples are scarce.

However, hyperspectral data inherently possess two challenging characteristics: the difficulty of annotation and significant data heterogeneity. First, regarding the difficulty

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of annotation, acquiring pixel-level ground truth for HSI is an extremely time-consuming and labor-intensive process that often requires expert domain knowledge. Consequently, existing methods struggle to strike a balance between sample efficiency and classification performance. Fully supervised methods typically depend on extensive labeled samples to train deep neural networks, limiting their feasibility in rapid response scenarios due to the aforementioned annotation costs [9]. Conversely, while zero-shot or unsupervised transfer methods alleviate the annotation burden, they often suffer from poor performance due to the absence of target-specific supervision to guide the selection of discriminative bands for specific land-cover classes [10]. Second, regarding data heterogeneity, HSIs acquired by different sensors inevitably exhibit inconsistencies in the number of bands, spectral ranges, and spatial resolutions [11]. These physical discrepancies make it difficult to align data in a unified feature space, thereby rendering many existing algorithms that assume fixed input dimensions or homogeneous distributions ineffective [12], [13]. Therefore, developing a few-shot heterogeneous band selection method capable of adapting to new sensor data with minimal supervision is of paramount importance.

Nevertheless, developing a robust few-shot heterogeneous band selection framework is nontrivial and cannot be achieved by a simple combination of existing techniques. Although researchers have attempted to address cross-scene issues using methods like CDIG [14], CDIRF [15], and CDWOASA [16], which primarily transfer knowledge by minimizing distributional discrepancies or optimizing statistical metrics, these approaches predominantly rely on shallow statistical characteristics. Crucially, they fail to capture the deep, nonlinear spectral dependencies essential for characterizing complex hyperspectral data. This limitation not only leads to the imprecise selection of domain-invariant bands but also renders them fundamentally incapable of handling heterogeneous data characterized by varying band numbers and spectral ranges. On the other hand, deep learning-based attempts such as MR-Selection [10] are constrained by structural limitations due to fixed output dimensions. Specifically, as the classifier is bound to the specific class count, MR-Selection requires training separate classifiers for each dataset, which hinders joint optimization and the automatic handling of new categories in unseen tasks. Consequently, three critical bottlenecks remain unaddressed: 1) the physical inconsistencies in spectral ranges and band counts across sensors make it difficult to construct a cohesive, unified feature representation for the agent; 2) existing methods lack a transferable sequential decision-making mechanism, failing to generalize search experiences from source tasks to rapidly adapt to unseen target tasks without retraining; and 3) the high variance of evaluation feedback derived from extremely scarce samples often destabilizes the optimization process, rendering standard reward functions unreliable. Addressing these specific challenges requires a unified framework capable of simultaneous feature alignment, robust policy transfer, and stability-enhanced evaluation.

In this article, we propose a novel few-shot heterogeneous hyperspectral band selection method, termed FHBS-UML. Diverging from existing deep learning-based methods that rely on static architectures tailored for specific datasets, this work enables rapid adaptation to unseen sensor data by establishing a transferable search mechanism. Since band selection is inherently a nondifferentiable combinatorial optimization problem lacking explicit supervision, the task is formulated as a sequential decision-making process within a meta-reinforcement learning (meta-RL) framework to optimize subsets across diverse source domains. To bridge the physical gap caused by inconsistent spectral ranges and band numbers, a heterogeneous HSI aligner (HSA) is designed. By utilizing a local–nonlocal transformer mechanism, the HSA captures invariant spectral dependencies to project diverse inputs into a unified feature space. Furthermore, domain shifts during few-shot evaluation often lead to high intraclass variance and unstable reward signals. To mitigate this, we introduce a stability-enhanced heterogeneous HSI evaluator. Unlike traditional prototypical networks relying on isotropic Euclidean distance, our evaluator employs a variance-sensitive metric based on adaptive covariance estimation to dynamically calibrate category confidence. Finally, to prevent the agent from being dominated by easier tasks during meta-training, an uncertainty-aware multiobjective optimization strategy is employed to balance the learning process across heterogeneous source tasks.

The main contributions of this article are summarized as follows.

1) To the best of our knowledge, this is the first work to address few-shot band selection for heterogeneous hyperspectral data. We propose a meta-RL-based general framework that captures the transferability of band selection policies across different sensors, enabling rapid adaptation to unseen tasks with varying configurations.
2) An HSA is designed to unify heterogeneous bands. By employing a local–nonlocal transformer encoder and feature alignment learning, HSA effectively aligns both the structural and feature distributions of HSIs with varying spectral ranges and band numbers, yielding cohesive band-level feature representations.
3) We propose a stability-enhanced heterogeneous HSI evaluator to mitigate the effects of large intraclass variance and inconsistent label sets, thereby significantly improving the stability and accuracy of band evaluation in heterogeneous scenarios.
4) An uncertainty-aware multiobjective meta-learning optimization strategy is proposed for highly generalized agent learning. This strategy leverages homoscedastic uncertainty to dynamically weigh multiple loss functions, ensuring that the agent learns a generalized policy capable of rapid adaptation to unseen target tasks.

II. Related Works

A. Single-Scene Band Selection

Traditional single-scene hyperspectral band selection methods are generally categorized into supervised, semi-supervised, and unsupervised approaches according to label availability [2]. As a representative shallow feature selection method, Peng et al. [17] proposed the max-dependency, max-relevance,

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and min-redundancy criteria, which provide an effective baseline by jointly considering band relevance and redundancy. With the advancement of deep learning, specialized end-to-end architectures have been further developed for band selection. Specifically, bandwise-independent binary architectures (BHCNNs) [18] incorporate learnable masking layers to explicitly identify informative bands. To reduce the dependence on extensive annotations, unsupervised and semi-supervised paradigms have also been explored, such as BS-Nets with L1-constrained spectral reconstruction [19] and semi-supervised 3-D-CNNs for adaptive spectral–spatial learning [20].

Beyond network design, recent studies have introduced advanced optimization frameworks to address the combinatorial complexity of band subset search. Feng et al. [21] pioneered the formulation of band selection as a Markov decision process (MDP) optimized via deep reinforcement learning (RL). More recently, evolutionary and heuristic algorithms have further improved search efficiency and stability; for example, Aravinth et al. [22] introduced the SRL-SOA algorithm to accelerate subset search, while Shang et al. [23] developed a hybrid framework combining Gray Wolf and Genetic algorithms to better balance exploration and exploitation. Graph-based methods such as DEHF [24] further model interband structural dependencies using dynamic graph enhancement, hierarchical feature fusion, and a multidimensional priority criterion, but they remain target-specific and are not designed for few-shot heterogeneous transfer. Although these methods differ in learning paradigm and optimization strategy, they are mainly designed for single-scene settings, where training and testing data share similar spectral distributions. As a result, the selected band subsets are often closely coupled with scene-specific spectral characteristics, which limits their transferability to new scenes and sensor configurations.

B. Cross-Scene Methods for Homogeneous Band Selection of HSIs

Generally, homogeneous cross-scene HSI data are acquired by the same type of sensor and therefore share identical spectral bands and wavelength ranges. Under this setting, the major challenge lies in spectral distribution shift caused by varying atmospheric conditions, illumination, and scene composition, which leads to inconsistent statistical characteristics of material signatures across different scenes. To mitigate this discrepancy and improve generalization performance, cross-scene methods, especially domain adaptation approaches [7], have been widely applied to homogeneous data. Ye et al. [14] proposed an information gain-based method, while other studies utilized interband redundancy analysis to preselect relatively independent bands [25]. In addition, algorithms such as RelieF have been extended into cross-domain versions [15].

More recently, dynamic optimization and self-representation learning have gained increasing attention. For example, a whale optimization-based method adopts a hybrid strategy to adapt to data distributions across domains [16], while advanced autoencoder architectures capture nonlinear spectral dependencies more effectively [26]. Subsequently, Sun et al. [27] proposed the cross-scene self-representative network (CSSRnet), which derives domain-invariant self-representative coefficients by enforcing distributional consistency across scenes. These methods have demonstrated the effectiveness of exploiting cross-scene shared information when source and target data are spectrally aligned. However, such alignment usually relies on the homogeneous assumption of consistent band correspondence, making these methods difficult to extend directly to heterogeneous hyperspectral data acquired from different sensors.

C. Cross-Scene Methods for Heterogeneous Band Selection of HSIs

Recent research has increasingly focused on processing heterogeneous data across different scenes. For classification tasks, domain adaptation techniques are widely adopted to align domain-specific features. Specifically, Peng et al. [12] employed deep transfer learning strategies to bridge the distribution gap between domains. To further address category shifts, discriminative cooperative alignment [8] has been proposed. However, these methods typically operate on full spectral bands and require abundant target samples to mitigate distributional discrepancies. Regarding heterogeneous band selection, current approaches predominantly rely on heuristic or evolutionary multiobjective optimization algorithms to handle data inconsistencies. For example, evolutionary strategies typically model tasks across different scenes as separate populations, dynamically adjusting search parameters, such as crossover and mutation probabilities, to adapt to specific data characteristics [28]. To address spectral mismatches, Zhang et al. [29] introduced a linear mapping strategy based on information entropy to filter varying spectral inputs before employing particle swarm optimization. Despite these advances, existing methods generally lack a mechanism to transfer search experience across heterogeneous sensors, necessitating computationally expensive task-specific adaptation or iterative reoptimization for each new sensor task. Furthermore, their reliance on sufficient supervision often compromises reliability in few-shot scenarios.

D. Meta-Learning Methods for Band Selection of HSIs

Meta-learning serves as a potent paradigm for handling heterogeneous and data-scarce environments [30]. For classification, approaches exemplified by model-agnostic meta-learning (MAML) [31] are designed to learn transferable representations, enabling models to quickly adapt to new tasks under scarce supervision. Similarly, semantic label embeddings have been employed to establish bidirectional correspondence between visual features and category semantics [32].

Regarding hyperspectral band selection, researchers have attempted to model unified meta-knowledge on the spectrum. Feng et al. [10] proposed MR-Selection, a two-stage meta-learning framework that aligns the update directions of shared meta-agents. Early zero-shot attempts based on RL explored the possibility of policy transfer in band selection, but their performance was often limited when spectral distributions

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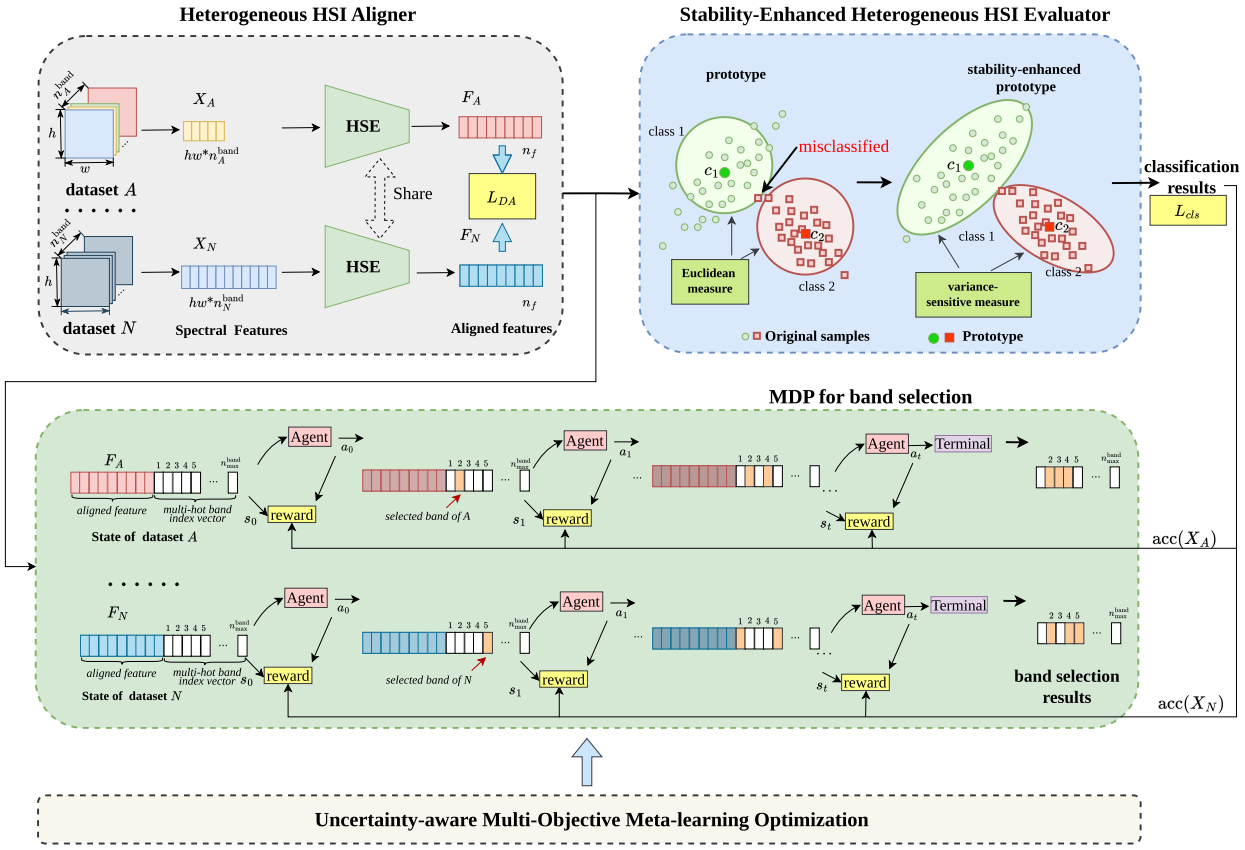


Fig. 1. Overview of the proposed FHBS-UML. The heterogeneous HSA (top-left) aligns spectral features from different domains, which are combined with band indices to form MDP states. The central MDP module then performs sequential band selection, guided by the stability-enhanced heterogeneous HSI evaluator (top-right) that assesses classification stability. The framework is optimized using an uncertainty-aware multiobjective meta-learning strategy (bottom) to facilitate effective transferability across heterogeneous datasets.

varied significantly across domains. Moreover, extending existing meta-learning frameworks to few-shot heterogeneous band selection remains challenging. In particular, when the number of labeled target samples is extremely limited, the evaluation feedback used to guide adaptation can become highly variable, which may reduce the stability of meta-optimization.

According to the above studies, establishing a stable and transferable band selection strategy remains a primary challenge in heterogeneous band selection, as most existing approaches require retraining for new tasks. In addition, due to physical inconsistencies across sensors, current methods generally lack a unified feature alignment mechanism to effectively bridge the spectral gap. Furthermore, the instability of evaluation resulting from scarce samples in the few-shot scenarios remains under-addressed, which limits the reliability of the optimization process.

III. Methodology

The overall architecture of the proposed FHBS-UML is illustrated in Fig. 1. The few-shot heterogeneous hyperspectral band selection and classification process is formalized as an MDP. Drawing inspiration from [21], we formulate the band search task as a sequence of decisions, where each state within the search space corresponds to a valid subset of bands. To address heterogeneity in the original feature spaces, an HSA is proposed to project the original spectra from the heterogeneous datasets into an aligned feature space, where the feature serves as the state in the MDP. In addition, a stability-enhanced heterogeneous HSI evaluator, designed as an adaptive variance-sensitive prototype for band subset classification, is pretrained as a band subset evaluator, with its output acting as the reward in the MDP. Finally, the capability of band selection for rapid transfer across heterogeneous datasets is further achieved through uncertainty-aware multiobjective meta-learning optimization. Once the RL agent is trained, informative band subsets can be obtained by fine-tuning the few-shot band selection task within this framework, allowing for direct derivation of classification results via the stability-enhanced heterogeneous HSI evaluator.

A. Modeling the Few-Shot Heterogeneous Band Selection and Classification as an MDP

The few-shot band selection task is constructed by sampling a dataset, referred to as an episode, from the source datasets and is formulated as an MDP defined by four elements: states, actions, transition functions, and rewards. Assuming the sampled HSI dataset contains a total of C_e classes, the aggregate sample set $\mathbf{X}$ is partitioned into three disjoint subsets. Specifically, a prototype set $\mathbf{X}_p$ is first designated to initialize the stability-enhanced heterogeneous HSI evaluator. Subsequently, a support set $\mathbf{X}_s$, containing n samples per class, is utilized to construct the MDP state, while the remaining

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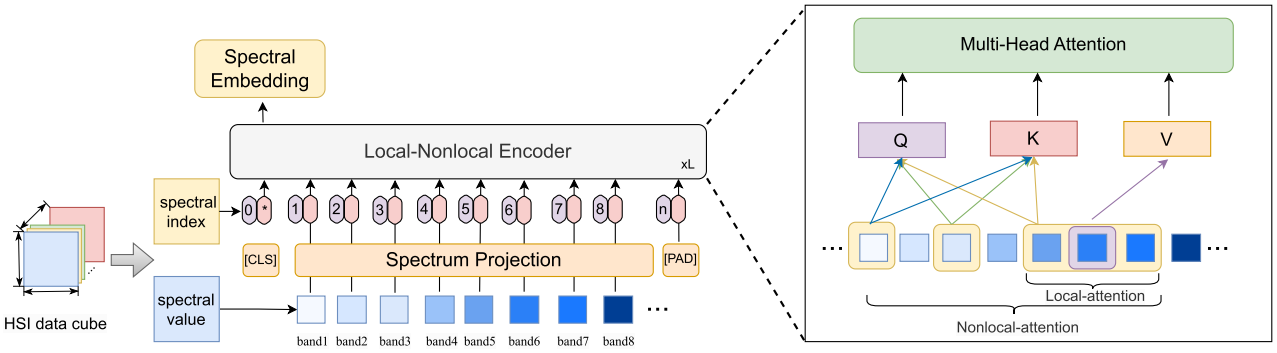


Fig. 2. Architecture of the local–nonlocal HSE, which processes hyperspectral data as sequences, combines linearly projected spectral reflectance values with spectral index embeddings to form token inputs. A specialized multihead attention mechanism integrates a local-attention module operating within a fixed window size w and a nonlocal-attention module utilizing LogSparse self-attention to capture both local spectral continuity and global cross-band dependencies.

m samples constitute the query set $\mathbf{X}_q$, which is employed to evaluate classification accuracy and compute the reward. Rather than optimizing band subsets independently for each dataset, this formulation enables the policy to reuse search experience across heterogeneous few-shot tasks.

1) *State:* At the current time step t, the state s_t is defined as $s_t = [f_t, b_t]$, where $[,]$ denotes the concatenation operation. The term $f_t \in \mathbb{R}^{n_f}$ refers to the aligned features of the currently selected bands derived from the HSA, and n_f denotes the feature dimension. The vector $b_t \in \mathbb{R}^{N_{\max}^{\text{band}}}$ is a multihot vector, where $N_{\max}^{\text{band}}$ denotes the maximum band capacity that the model can accommodate. $N_{\max}^{\text{band}}$ was set to 256 as a fixed padding capacity to unify the state dimension across heterogeneous datasets, which is sufficient to cover all adopted datasets. Elements corresponding to the selected bands are set to 1, while elements corresponding to unselected valid bands are set to 0 in b_t, entries beyond the actual band number are padded with zeros, and excluded from the valid action space. Existing homogeneous band selection methods [10], [21] rely solely on the use of multihot vectors for state modeling. However, this approach overlooks the spectral reflectance characteristics of the bands, resulting in insufficient information that renders it inadequate for heterogeneous band selection tasks.

2) *Action:* Action $a_t \in \mathcal{A}$ is to select a band at the current time step t, where $\mathcal{A}$ denotes the action space consisting of candidate bands. The probability of selecting a band is determined by the agent's strategy.

3) *State Transition Function:* Upon the execution of an action at time step t, the current state s_t transitions to the subsequent state s_{t+1}. This process is determined by the state transition function T. The aligned feature f_{t+1} in the new state s_{t+1} is obtained by inferring the support set $\mathbf{X}_s$ with the HSA after updating the selected bands. The update of the multihot vector b_{t+1} is achieved by setting the element corresponding to the newly selected band to 1. When the number of selected bands reaches the preset value K_b, the episode concludes, i.e., the state transitions to Terminal. The state transition function T can be represented by the following formula:

$$s_{t+1} = \begin{cases} \text{Terminal}, & \text{if Num}\left(b_{t+1}\right) = K_b \\ \left[f_{t+1}, b_{t+1}\right], & \text{otherwise} \end{cases} \tag{1}$$

where K_b denotes the target number of selected bands.

4) *Reward:* In FHBS-UML, intermediate rewards are set to 0 until the selected subset size equals the predetermined K_b. When the number of selected bands reaches K_b, the classification accuracy of the query set $\mathbf{X}_q$, evaluated by the stability-enhanced heterogeneous HSI evaluator with the selected bands, is used as a reward. We define the reward function as follows:

$$r_t = \begin{cases} \text{acc}\left(\mathbf{X}_q\right), & \text{if Num}\left(b_{t+1}\right) = K_b \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where Num$(\cdot)$ denotes the count of nonzero elements (1s) in the vector, while acc$(\cdot)$ computes the classification accuracy on the query set $\mathbf{X}_q$ using the stability-enhanced heterogeneous HSI evaluator.

B. Heterogeneous HSA

Domain alignment is a crucial step for cross-scene HSI analysis. Existing methods [11] require designing a separate network for each dataset, which limits them to handling only a fixed set of data dimensions. This approach is therefore unsuitable for the heterogeneous band selection problem, where the number of bands varies across different datasets. In this section, HSA is designed to align heterogeneous HSI data in terms of feature space alignment and feature distribution alignment. Specifically, fixed-dimensional encoders are constrained by predefined input structures, while feature distribution matching alone cannot explicitly handle variations in band numbers and spectral structures. To address these limitations, the local–nonlocal heterogeneous HSI encoder (HSE) performs variable-length spectral encoding for feature space alignment, and feature alignment learning is employed during training to reduce distribution discrepancies across domains. The architecture of the proposed HSA is shown in the upper left part of Fig. 1.

Traditional feature extraction networks employing convolutional neural networks and recurrent neural networks [33], [34] have limitations in cross-band relationships extraction due to their focus on local relationships. The Transformer model [35], due to its self-attention mechanism, is highly effective

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at discerning global relationships between tokens, making it particularly suitable for extracting cross-band relationships in hyperspectral data. In the HSA, hyperspectral data are initially encoded as sequences along the band dimension. Each band's spectral reflectance values are represented as patch tokens and undergo a linear projection to produce 1-D token embeddings, which are subsequently input into the transformer encoder. The sequential position of each band provides positional encoding. HSA's ability to handle variable-length sequence inputs while extracting spectral relationships further facilitates structural alignment of heterogeneous hyperspectral data. Previous studies have noted that hyperspectral data exhibit significant local correlations, meaning that the spectral reflectance values are continuous along the spectral curve. In addition, the nonlocal correlations within hyperspectral data cannot be overlooked. The inherent lack of sensitivity to local context in the point-wise dot product self-attention of standard Transformers renders the model vulnerable to anomalies in heterogeneous HSI data.

Inspired by [36], a local–nonlocal HSE is designed, as shown in Fig. 2. The proposed local–nonlocal HSE computes self-attention in a novel way by dividing the attention mechanism into two components: local-attention and nonlocal-attention. Instead of applying the query matrix Q of each value matrix V across all bands, the local-attention operates within a window of size w, where w denotes the local attention window size. The nonlocal interaction is realized via LogSparse self-attention. In this scheme, each cell attends strictly to its own position and a subset of predecessors sampled with an exponentially growing stride. Specifically, for any layer index r and token position l, the index set I_l^r is defined as

$$I_l^r = \begin{cases} \{l-w, \ldots, l-1, l, l+1, \ldots, l+w\}, \\ \qquad\qquad \text{if } l-w \geq 1 \text{ and } l+w \leq L \\ \left\{l - 2^{\lfloor \log_2(l-w) \rfloor}, l - 2^{\lfloor \log_2(l-w) \rfloor - 1}, \ldots, l - 2^0 \right. \\ \qquad \left. \ldots, l + 2^{\lfloor \log_2(l-w) \rfloor} \right\}, \\ \qquad\qquad \text{otherwise} \end{cases} \tag{3}$$

where L denotes the sequence length, $\lfloor \cdot \rfloor$ denotes the floor operation, and I_l^r indicates the set of indices of the cells that cell l can attend to during the computation from the rth layer to the $(r+1)$th layer.

After aligning the feature space, the method for homogeneous cross-scene band alignment can be applied to achieve feature distribution alignment. In this work, the MMD loss [37] is employed for feature-level alignment. To compute the MMD loss pairwise among N_d source domains, feature sets extracted by the transformer encoder from each domain are denoted as $X_1, X_2, \ldots, X_{N_d}$. For domains i and j, where $i \neq j$ and $i, j \in \{1, 2, \ldots, N_d\}$, the MMD loss, MMD_{ij}, is calculated as follows:

$$\text{MMD}_{ij}^2 = \left\| \frac{1}{n_i} \sum_{x \in X_i} \phi(x) - \frac{1}{n_j} \sum_{x \in X_j} \phi(x) \right\|^2 \tag{4}$$

where $\phi(x)$ represents the feature mapping function, that is, HSE transforms input data into a new feature space, and n_i and n_j are the sample numbers in domains i and j, respectively.

The pairwise MMD loss across all N_d source domains is computed by evaluating MMD_{ij} for each unique pair (i, j) where $i < j$. This is formally expressed as

$$L_{DA} = \sum_{i=1}^{N_d - 1} \sum_{j=i+1}^{N_d} \text{MMD}_{ij}^2. \tag{5}$$

The introduction of the MMD loss has significantly reduced the discrepancies in the feature distributions learned from the source domains, thereby facilitating feature distribution alignment of heterogeneous hyperspectral data.

C. Stability-Enhanced Heterogeneous HSI Evaluator

Evaluating the classification performance of band subsets within inconsistent ground category sets presents a substantial challenge in the field of band selection. Prototypical networks [38] are specialized machine learning models tailored for few-shot classification tasks, wherein they construct a metric space that facilitates classification by calculating distances to prototype representations of each class. However, standard prototypical evaluators usually rely on isotropic distance metrics, which provide limited modeling of class-dependent variations and lead to unstable reward estimation and classification results in heterogeneous few-shot episodes. To address this issue, this section introduces a variance-sensitive prototypical network that incorporates class-dependent covariance information to improve the robustness of classification results under such conditions.

To learn suitable prototype representations for few-shot band evaluation, we mimic the few-shot classification task by sampling episodic minibatches. The HSA module computes the prototype representation of each class using the embedding function $f_\phi : \mathbb{R}^D \to \mathbb{R}^M$, where f_ϕ denotes the HSA embedding function parameterized by learnable parameters ϕ, and D and M denote the input and output dimensions of the network, respectively.

Specifically, model updates are performed using gradients derived from episodes, which are formed by randomly drawing categories from the source training data. For each selected class, samples are partitioned into a support set and a query set. These sets are then processed by the network to obtain feature embeddings. Let S_k be the set of samples for the kth class within the support set, where $|S_k| = N_s^k$, as defined below

$$S_k = \{(x_i, y_i)\}_{i=1}^{N_s^k} \tag{6}$$

where x_i represents the ith sample in the kth class sample set, while y_i denotes its corresponding label. Subsequently, the class prototype is computed using the embedding function and the averaging operation

$$c_k = \frac{1}{|S_k|} \sum_{(x_i, y_i) \in S_k} f_\phi(x_i). \tag{7}$$

To improve the robustness of prototype-based evaluation under heterogeneous few-shot episodes, we further introduce a variance-sensitive distance metric. Within the metric space formed by the embedding function f_ϕ, the distance between the query sample and each class prototype is calculated, and

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then softmax is used for classification. The probability of x being classified as the kth class can be calculated as follows:

$$p_\phi(y=k|x)=\frac{\exp\left(-md_k\left(f_\phi(x),c_k\right)\right)}{\sum_{c=1}^{C_e}\exp\left(-md_c\left(f_\phi(x),c_c\right)\right)} \tag{8}$$

where $md_k(\cdot)$ represents the calculation of variance-sensitive distance, $\exp(\cdot)$ represents the exponential function, and C_e represents the number of classes within an episode. Specifically, $md_k(\cdot)$ is described as follows:

$$md_k(x,y)=\frac{1}{2}(x-y)^T\mathbf{Q}_k^{-1}(x-y) \tag{9}$$

where $\mathbf{Q}_k$ is a covariance matrix corresponding to the kth class in the current episode. Due to the unavailability of $\mathbf{Q}_k$ in advance, the embedding features of the support samples in the current episode are used to estimate $\mathbf{Q}_k$

$$\mathbf{Q}_k=\omega_k\mathbf{\Sigma}_k+(1-\omega_k)\mathbf{\Sigma}_e+\mathbf{I} \tag{10}$$

where $\mathbf{\Sigma}_k$ denotes the intraclass covariance matrix of the kth class, and $\mathbf{\Sigma}_e$ denotes the episode-level covariance matrix estimated from the support samples of all classes in the current episode. $\mathbf{I}$ represents the identity matrix. ω_k is used to achieve a tradeoff between $\mathbf{\Sigma}_k$ and $\mathbf{\Sigma}_e$, where $\omega_k=|S_k|/(|S_k|+1)$. As the number of support samples for the kth class increases, $\mathbf{Q}_k$ places greater emphasis on the intraclass covariance matrix $\mathbf{\Sigma}_k$. Moreover, unlike $\mathbf{\Sigma}_k$, which only utilizes the support samples of the kth class, $\mathbf{\Sigma}_e$ exploits the support samples of all classes to estimate the episode-level covariance matrix.

We incorporate a distance regularization term to improve the separability of the metric space. This component works by compressing the distance between the query sample and its ground-truth prototype, while simultaneously increasing the separation from the nearest nontarget prototype. The term $\mathcal{L}_{reg}$ is formulated as the ratio

$$\mathcal{L}_{reg}=\frac{1}{N_q}\sum_{(x,y)\in\mathcal{Q}}\frac{md_y\left(f_\phi(x),c_y\right)}{\min_{j\neq y}md_j\left(f_\phi(x),c_j\right)} \tag{11}$$

where $\mathcal{Q}$ denotes the query set in the current episode, $N_q=|\mathcal{Q}|$ denotes the number of query samples, c_y denotes the class prototype of the ground-truth class y, and the denominator represents the distance to the closest prototype of a different class. Minimizing this ratio effectively enforces intraclass compactness and interclass separability.

Finally, the total objective function combines the classification loss and the distance regularization term. To improve readability, the total loss is rewritten by explicitly separating the two components as follows:

$$\begin{aligned}\mathcal{L}&=\mathcal{L}_{cls}+\lambda_{\text{reg}}\mathcal{L}_{reg}\\ \mathcal{L}_{cls}&=\frac{1}{N_q}\sum_{(x,y)\in\mathcal{Q}}\left[md_y\left(f_\phi(x),c_y\right)\right.\\ &\qquad\left.+\log\sum_{c=1}^{C_e}\exp\left(-md_c\left(f_\phi(x),c_c\right)\right)\right]\\ \mathcal{L}_{reg}&=\frac{1}{N_q}\sum_{(x,y)\in\mathcal{Q}}\frac{md_y\left(f_\phi(x),c_y\right)}{\min_{j\neq y}md_j\left(f_\phi(x),c_j\right)}\end{aligned} \tag{12}$$

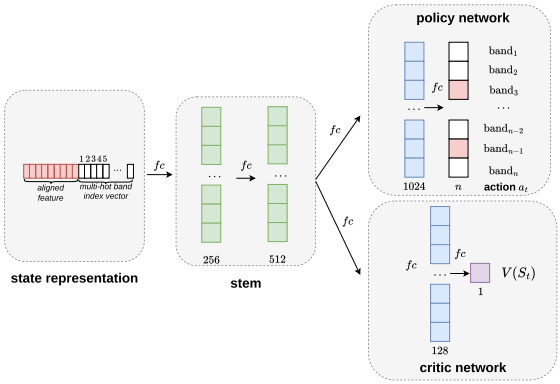


Fig. 3. Structure of the A2C-RL agent. The input state representation, comprising aligned features and a multihot band index vector, is processed through a shared stem network of fully connected layers. The architecture then branches into two components: a policy network (f_{θ_a}) that outputs the action a_t for band selection, and a critic network (f_{θ_v}) that estimates the state value $V(S_t)$.

where λ_{reg} is the adjustment factor for the distance regularization term. By effectively correcting classification confidence using variance and introducing an intraclass constraint loss as a regularization term, the evaluator can provide more stable and discriminative feedback for heterogeneous few-shot band evaluation.

D. Uncertainty-Aware Multiobjective Meta-Learning Optimization

The core objective of meta-RL is to train an agent with robust generalization capabilities, enabling rapid adaptation to new tasks by leveraging historical experience from source datasets. Previous work [10] utilized MAML [31] to achieve this transfer. However, standard MAML treats all source tasks equally, which is inadequate for heterogeneous band selection. In this setting, tasks constructed from different sensors can exhibit distinct difficulty levels and noise levels, which are regarded as task uncertainty in this work. Ignoring these uncertainty-related variations can lead to suboptimal convergence [39]. To address this, we propose an uncertainty-aware multiobjective meta-learning optimization method, which dynamically reweights tasks based on their intrinsic uncertainty and reduces the influence of unstable or noisy feedback during the meta-update.

1) A2C-Based Optimization: We employ the advantage actor–critic (A2C) algorithm as the base optimizer. As illustrated in Fig. 3, the agent comprises a policy network f_{θ_a} and a critic network f_{θ_v}. For a specific few-shot task $\mathcal{T}_i$, the agent interacts with the environment to generate a trajectory $\mathcal{D}=\{(s_j,a_j,r_j)\}_{j=1}^{h}$, where h denotes the trajectory length. Here, h dynamically equals the target number of selected bands, i.e., $h=K_b$, since one band is selected at each decision step. The policy network is updated to maximize the expected reward. The policy loss is defined as

$$\mathcal{L}_{\mathcal{T}_i}^{policy}(\theta_a)=-\sum_{j=1}^{h}\log\pi\left(a_j|s_j;\theta_a\right)A\left(s_j,a_j\right) \tag{13}$$

where $\pi(a_j|s_j;\theta_a)$ denotes the action probability produced by the policy network, $A(s_j,a_j)=R_j-V(s_j;\theta_v)$ represents the

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advantage function, and $R_j = \sum_{k=0}^{h-j} \gamma^k r_{j+k}$ is the discounted cumulative reward with discount factor γ. The critic network aims to minimize the prediction error of the value function, using the mean squared error (mse) loss

$$\mathcal{L}_{\mathcal{T}_i}^{critic}(\theta_v) = \sum_{j=1}^{h} \left(R_j - V\left(s_j; \theta_v\right)\right)^2. \tag{14}$$

2) Uncertainty-Aware Meta-Update: In the meta-learning process, we distinguish between task-specific updates in the inner loop and meta-parameter updates in the outer loop. In the inner loop, for each task $\mathcal{T}_i$, the agent performs a tentative update using the support set trajectory $\mathcal{D}_{\mathcal{T}_i}^{sup}$

$$\theta_a' = \theta_a - \alpha \nabla_{\theta_a} \mathcal{L}_{\mathcal{T}_i}^{policy}(\theta_a), \quad \theta_v' = \theta_v - \alpha \nabla_{\theta_v} \mathcal{L}_{\mathcal{T}_i}^{critic}(\theta_v) \tag{15}$$

where α is the inner learning rate. In the outer loop, the agent is evaluated on the query set trajectory $\mathcal{D}_{\mathcal{T}_i}^{que}$ using the updated parameters θ_a', θ_v'.

Standard approaches simply average the losses across tasks, which is typically formulated as

$$\mathcal{L}_{standard}(\theta) = \frac{1}{N_t} \sum_{i=1}^{N_t} \lambda_i \mathcal{L}_{\mathcal{T}_i}^{total}\left(\theta_a', \theta_v'\right) \tag{16}$$

where N_t denotes the number of tasks in one meta-batch, and λ_i denotes the weight of task $\mathcal{T}_i$ in the standard formulation. However, tasks characterized by high uncertainty and unstable rewards should contribute less to the meta-update. Inspired by [39], we model the task-dependent uncertainty using homoscedastic uncertainty learning. We introduce a learnable uncertainty parameter σ_i for each task $\mathcal{T}_i$. The multiobjective meta-loss function combines the policy and critic losses, formulated as

$$\mathcal{L}_{meta}(\theta, \boldsymbol{\sigma}) = \frac{1}{N_t} \sum_{i=1}^{N_t} \left(\frac{1}{2\sigma_i^2} \mathcal{L}_{\mathcal{T}_i}^{total}\left(\theta_a', \theta_v'\right) + \log \sigma_i \right) \tag{17}$$

where $\mathcal{L}_{\mathcal{T}_i}^{total} = \mathcal{L}_{\mathcal{T}_i}^{policy} + \mathcal{L}_{\mathcal{T}_i}^{critic}$ denotes the joint loss on the query set. $\boldsymbol{\sigma} = \{\sigma_1, \ldots, \sigma_{N_t}\}$ represents the set of learnable task uncertainty parameters. The term $1/2\sigma_i^2$ acts as a dynamic weight: when a task has high uncertainty with a large σ_i, its contribution to the gradient is attenuated. The regularization term $\log \sigma_i$ prevents σ_i from increasing infinitely.

Finally, the meta-parameters θ and task uncertainty parameters $\boldsymbol{\sigma}$ are updated jointly via gradient descent on $\mathcal{L}_{meta}$

$$\theta \leftarrow \theta - \beta \nabla_\theta \mathcal{L}_{meta}, \quad \boldsymbol{\sigma} \leftarrow \boldsymbol{\sigma} - \beta \nabla_{\boldsymbol{\sigma}} \mathcal{L}_{meta} \tag{18}$$

where β is the outer learning rate. This mechanism allows the agent to automatically balance the contribution of heterogeneous tasks, focusing on stable, high-quality feedback during training.

3) Fine-Tuning on Target Data: When adapting to a new target dataset with T classes and n labeled samples per class, the few-shot labeled target training set is denoted as $\mathbf{X}_{target}$. Following the few-shot protocol, $\mathbf{X}_{target}$ is used as the support set $\mathbf{X}_s$ for target adaptation, while the remaining target samples are used for final evaluation. Building upon the meta-trained parameters, the agent interacts with $\mathbf{X}_s$ to sample trajectories and performs a few steps of A2C updates, similar to the inner loop process. This rapid fine-tuning process aligns the agent's policy with the specific spectral characteristics of the target sensor, yielding a robust band selection strategy with minimal computational cost.

TABLE I
SUMMARY OF THE ADOPTED HYPERSPECTRAL DATASETS. BANDS ARE REPORTED AS ORIGINAL/USED AFTER NOISY OR WATER-ABSORPTION BAND REMOVAL

Dataset	Sensor	Spec. Range (μm)	Bands	Spatial Res.
Botswana	Hyperion	0.40–2.50	242/145	30 m
KSC	AVIRIS	0.40–2.50	224/176	18 m
Houston	CASI-1500	0.38–1.05	144/144	2.5 m
Washington DC	HYDICE	0.40–2.40	210/191	∼2.8 m
IP	AVIRIS	0.40–2.50	224/200	20 m
SA	AVIRIS	0.40–2.50	224/204	3.7 m
PU	ROSIS	0.43–0.86	115/103	1.3 m

IV. EXPERIMENTAL RESULTS AND ANALYSIS

To comprehensively substantiate the efficacy and robustness of the proposed FHBS-UML methodology, a series of extensive experiments was conducted. Specifically, these experiments involve a total of 7 hyperspectral datasets, namely, Indian Pines (IP), Salinas (SA), Pavia University (PU), Botswana, Kennedy Space Center (KSC), University of Houston, and Washington DC Mall (WDC).

Table I summarizes the sensor type, spectral range, number of bands, and spatial resolution of the adopted datasets, where Bands denotes the original number of spectral bands and the number of bands actually used after removing noisy or water-absorption bands, respectively.

Following standard conventions in few-shot classification, the experimental data is categorized into source domain data for meta-training and target domain data for evaluation. Specifically, for the main experiments on the target datasets, including IP, SA, and PU, we randomly select five labeled samples per class to compose the training set. These samples serve as the support set for domain adaptation, while the remaining samples are retained for testing to conduct a comprehensive performance assessment. To facilitate knowledge transfer and align the support set configuration with the target domain, an episodic training strategy is employed on the source datasets for a total of 10 000 training episodes. During this meta-training phase, the 5 support samples and 15 query samples per class are dynamically resampled in each episode. To ensure statistical stability and mitigate the variance induced by random sampling, the entire experimental procedure is repeated for ten independent runs using different random seeds. In each run, FHBS-UML performs meta-training on the source domains, fine-tuning on the randomly sampled target support set, and testing on the remaining target samples. The results reported in the comparison tables represent the mean and standard deviation across these complete runs.

A. Experimental Setups

To more convincingly demonstrate the effectiveness of our algorithm in various aspects, 11 existing band selection

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algorithms are chosen for performance comparison, including MRMR [17], BS-Nets [19], DDCNN [40], BHCNN-CFL [18], CAE-UBS [26], RLSBS [21], SRL-SOA [22], GWOGA [23], DEHF [24], EMAC-BS [29], and CSSRnet [27]. These methods cover representative classical and recent SOTA band selection approaches.

The single-scene methods include classical algorithms (MRMR), reinforcement-learning-based methods (RLSBS), heuristic-based approaches (SRL-SOA, GWOGA), and deep learning-based models (BS-Nets, DDCNN, BHCNN-CFL, CAE-UBS, and DEHF). For the MRMR method, we apply a forward selection scheme that prioritizes minimizing redundancy while maximizing relevance, employing a support vector machine to evaluate the selected bands. For RLSBS, we followed the original RL-based band selection framework and parameter settings. For SRL-SOA, parameters are set to the default values suggested in the original article. For GWOGA, the population size and maximum iterations are set to 30 and 100, respectively, following the standard configuration for its hybrid optimization strategy. Regarding the deep learning-based single-scene methods: BS-Nets [19] is configured with a learning rate of 0.001, 500 training epochs, and a regularization coefficient of 0.02. DDCNN [40] follows the network structure and hyperparameters described in [40]. For BHCNN-CFL and CAE-UBS, we adopt the architectures designed in their original studies. For DEHF, we followed its original dynamic graph enhancement and hierarchical feature fusion framework. In heterogeneous experiments, band dimensions were aligned by zero-padding, and padded bands were masked out during graph construction and final band selection. The cross-scene methods include EMAC-BS, CSSRnet, and our proposed method. These methods are evaluated under a cross-scene setting (training on source, testing on target). For EMAC-BS, the evolutionary parameters and knowledge transfer coefficients are initialized based on its default multitasking framework settings. For CSSRnet, we followed its original cross-scene self-representation framework and training settings.

The proposed FHBS-UML framework was implemented in PyTorch using Python under CUDA 12.9. All experiments were conducted on a workstation equipped with an AMD Ryzen 9 7950X 16-Core Processor and an NVIDIA GeForce RTX 4090 GPU. To improve the training stability in the few-shot setting, the overall optimization was decoupled into feature representation learning and meta-RL policy learning. For the HSA and the Stability-Enhanced Heterogeneous HSI Evaluator, the Adam optimizer was used with a learning rate of 1×10^{-3}. In the local–nonlocal encoder, the local attention window size (w) was set to 2, and the distance regularization factor (λ_{reg}) was set to 0.1.

For the meta-RL agent, the Actor and Critic networks were optimized using RMSprop. The inner loop adaptation learning rate (α) and the outer loop meta-learning rate (β) were both set to 1×10^{-4}. The number of inner update steps was set to 5. In the reinforcement learning (RL) process, the discount factor (γ) used for calculating the discounted cumulative reward was set to 0.999, and the entropy regularization coefficient was set to 0.01.

B. Single-Scene Performance

In this section, to preliminarily verify the effectiveness of our proposed network architecture, we first validate its classification performance under a standard single-scene setting. Distinct from few-shot learning frameworks, the training and testing sets in this experiment are derived from a single dataset. Quantitative performance is evaluated using three standard metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). The comprehensive results of all comparison algorithms across the three datasets are summarized in Table II.

We evaluated the performance on the IP, SA, and PU datasets, partitioning samples into 5%, 1%, and 3% training sets (with the remainder for testing) and selecting 20, 15, and 20 bands, respectively. As detailed in Table II, our FHBS-UML consistently outperforms other methods across all datasets. On the IP dataset, shallow models like BS-Nets and MRMR exhibited limited performance due to their reliance on handcrafted features and lack of joint optimization. While deep learning approaches like DDCNN and BHCNN-CFL improved results through spatial feature extraction, our method achieved the best performance with an average OA of 94.9%, surpassing the second-best RLSBS by 5.1%. For the SA dataset, comparison methods encountered performance bottlenecks despite the large homogeneous regions; specifically, the second-ranked RLSBS reached 96.6% OA while others remained below 96%. In contrast, our method achieved better average performance, outperforming the closest competitor by 2.7%. A similar trend was observed on the PU dataset, where FHBS-UML again yielded the best results with an OA of 98.9%, AA of 98.5%, and Kappa of 98.3%, improving these metrics by 5.1%, 5.9%, and 5.0%, respectively, compared to the suboptimal RLSBS method. This superior performance is primarily attributed to the adaptive variance-sensitive prototype in the stability-enhanced heterogeneous HSI evaluator, which effectively handles high intraclass variance, and the sequential decision-making process in meta-RL that captures deep nonlinear spectral dependencies.

C. Cross-Scene Performance

To evaluate the generalization capability and robustness of the proposed algorithm under label-scarce conditions, comparative experiments were conducted using a five-shot setting on three target datasets: IP, SA, and PU. We adopted a strict source-target separation strategy where Botswana, KSC, Houston, and Washington DC served as source domains for meta-training, while the target datasets were strictly isolated for testing. During meta-training, 15 samples per class were used as the query set to optimize the model. For the target domains, we followed a standard few-shot protocol by randomly selecting only five labeled samples per class as the support set for adaptation, retaining all remaining samples for testing to objectively reflect full-map classification performance. Regarding band selection, the number of selected bands was set to 15 for SA and 20 for both IP and PU.

To enable a fair comparison across datasets with heterogeneous spectral dimensions, we applied the same adaptation

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TABLE II
COMPARISON OF BAND SELECTION METHODS IN A SINGLE-SCENE SETTING

Category	Method	Indian Pines			Salinas			Pavia University		
		OA(%)	AA(%)	Kappa×100	OA(%)	AA(%)	Kappa×100	OA(%)	AA(%)	Kappa×100
Single Scene Methods	MRMR [17]	74.6 ± 0.6	66.9 ± 2.4	70.8 ± 0.7	86.8 ± 0.6	84.8 ± 0.6	85.2 ± 0.6	82.5 ± 0.5	76.4 ± 0.3	76.2 ± 0.6
	BS-Nets [19]	72.0 ± 0.9	64.0 ± 1.9	67.9 ± 1.0	85.3 ± 0.8	82.5 ± 1.3	83.6 ± 1.0	81.9 ± 0.9	74.4 ± 2.0	75.1 ± 1.4
	DDCNN [40]	80.5 ± 0.3	74.3 ± 0.2	77.6 ± 0.3	90.3 ± 0.4	88.8 ± 0.4	89.2 ± 0.5	86.1 ± 0.5	80.8 ± 1.5	81.0 ± 0.9
	BHCNN-CFL [18]	86.0 ± 0.8	79.5 ± 0.5	82.4 ± 0.8	94.9 ± 0.4	92.2 ± 0.6	92.8 ± 0.5	90.5 ± 0.6	84.9 ± 0.8	86.1 ± 0.5
	CAE-UBS [26]	81.9 ± 1.0	75.6 ± 1.8	79.0 ± 1.1	93.3 ± 0.5	91.1 ± 0.7	92.4 ± 0.5	92.0 ± 0.4	88.5 ± 0.9	88.8 ± 0.5
	RLSBS [21]	89.8 ± 1.1	86.6 ± 1.8	87.4 ± 1.3	96.6 ± 0.6	94.5 ± 0.7	94.9 ± 0.7	93.8 ± 0.5	92.6 ± 0.9	93.3 ± 0.5
	SRL-SOA [22]	82.2 ± 1.0	74.9 ± 0.8	78.7 ± 0.5	92.3 ± 0.7	90.7 ± 0.9	90.9 ± 0.5	87.7 ± 0.7	82.4 ± 0.7	82.8 ± 0.8
	GWOGA [23]	82.6 ± 0.6	78.4 ± 1.0	79.5 ± 0.7	93.7 ± 0.5	90.3 ± 0.8	90.5 ± 0.5	88.2 ± 0.6	82.8 ± 0.9	84.4 ± 0.7
	DEHF [24]	86.3 ± 1.4	81.7 ± 1.2	82.4 ± 1.1	94.8 ± 0.7	92.5 ± 0.8	93.1 ± 0.9	91.3 ± 1.1	88.3 ± 1.0	89.1 ± 1.3
Cross-Scene Methods	EMAC-BS [29]	79.7 ± 0.4	78.5 ± 0.4	77.2 ± 0.5	93.3 ± 0.5	91.4 ± 0.7	89.6 ± 0.6	86.8 ± 0.5	85.7 ± 0.6	84.2 ± 0.5
	CSSRnet [27]	81.4 ± 0.4	80.8 ± 0.4	78.2 ± 0.5	95.4 ± 0.2	92.8 ± 0.4	91.4 ± 0.3	89.8 ± 0.4	89.5 ± 0.5	85.7 ± 0.4
	Ours	**94.9 ± 0.5**	**94.0 ± 1.1**	**93.1 ± 0.6**	**99.3 ± 0.2**	**99.0 ± 0.3**	**98.7 ± 0.2**	**98.9 ± 0.2**	**98.5 ± 0.3**	**98.3 ± 0.4**

TABLE III
COMPARISON OF BAND SELECTION METHODS UNDER FIVE-SHOT CONDITIONS IN CROSS-SCENE SETTING

Category	Method	Indian Pines			Salinas			Pavia University		
		OA(%)	AA(%)	Kappa×100	OA(%)	AA(%)	Kappa×100	OA(%)	AA(%)	Kappa×100
Single Scene Methods	MRMR [17]	52.2 ± 2.1	59.4 ± 1.7	46.4 ± 2.1	78.3 ± 2.2	85.9 ± 2.3	75.9 ± 3.7	66.6 ± 3.0	74.6 ± 1.7	57.5 ± 3.5
	BS-Nets [19]	38.2 ± 2.6	41.4 ± 1.4	37.0 ± 2.6	63.9 ± 3.3	69.5 ± 2.3	60.6 ± 3.9	48.7 ± 3.7	59.5 ± 2.0	41.5 ± 3.4
	DDCNN [40]	50.7 ± 2.5	61.6 ± 3.2	44.7 ± 2.0	80.4 ± 3.3	87.5 ± 3.9	78.1 ± 3.8	66.3 ± 3.1	78.3 ± 2.9	58.2 ± 3.1
	BHCNN-CFL [18]	57.9 ± 3.3	73.0 ± 2.5	53.7 ± 2.9	71.3 ± 3.8	77.0 ± 4.0	68.1 ± 2.2	58.8 ± 2.9	70.4 ± 2.7	50.2 ± 2.4
	CAE-UBS [26]	50.3 ± 2.7	61.9 ± 3.1	44.3 ± 2.3	83.0 ± 2.4	88.4 ± 2.2	80.9 ± 2.8	75.5 ± 3.3	81.0 ± 2.9	67.8 ± 3.6
	RLSBS [21]	55.1 ± 2.2	58.9 ± 3.5	45.6 ± 2.0	82.9 ± 3.3	89.5 ± 1.9	80.2 ± 3.6	73.2 ± 2.3	83.4 ± 4.7	67.5 ± 3.3
	SRL-SOA [22]	49.0 ± 2.8	52.7 ± 2.8	42.6 ± 2.9	80.4 ± 2.2	85.2 ± 3.9	78.5 ± 3.4	67.3 ± 3.7	74.4 ± 3.0	58.4 ± 3.1
	GWOGA [23]	58.4 ± 3.3	64.4 ± 3.3	52.4 ± 2.9	78.3 ± 2.6	85.3 ± 3.3	75.2 ± 3.8	71.4 ± 3.9	77.3 ± 2.3	66.2 ± 4.2
	DEHF [24]	56.4 ± 3.1	63.8 ± 2.9	51.9 ± 3.4	78.9 ± 3.2	86.1 ± 2.5	75.6 ± 2.9	74.1 ± 2.7	79.2 ± 2.5	70.6 ± 2.8
Cross-Scene Methods	EMAC-BS [29]	55.0 ± 3.7	63.4 ± 3.4	48.4 ± 3.4	76.4 ± 3.4	83.0 ± 3.8	73.0 ± 2.4	73.0 ± 2.3	76.0 ± 2.0	67.2 ± 3.0
	CSSRnet [27]	56.9 ± 3.2	67.9 ± 3.5	52.9 ± 2.9	79.9 ± 3.5	88.4 ± 3.3	75.7 ± 2.3	74.1 ± 2.3	80.0 ± 3.2	70.3 ± 2.9
	Ours	**70.4 ± 1.6**	**78.4 ± 1.4**	**63.6 ± 1.2**	**89.7 ± 1.5**	**94.7 ± 1.2**	**87.6 ± 1.6**	**82.9 ± 2.3**	**87.5 ± 1.8**	**76.5 ± 2.4**

protocol to all baselines. In this protocol, each dataset was zero-padded to 256 spectral bands, matching $N_{\max}^{\text{band}} = 256$ in FHBS-UML, and the padded positions were masked during both band selection and evaluation to prevent any method from selecting or being evaluated on uninformative bands. This preprocessing only aligned the input dimensions and did not alter the original learning objectives or optimization strategies of the baselines.

The quantitative results, including OA, AA, and the kappa coefficient, are summarized in Table III. As shown in Table III, the proposed method consistently outperforms all 11 state-of-the-art baselines across the three datasets. Specifically, single-scene methods generally exhibit limited performance because the scarcity of training samples leads to overfitting or insufficient feature representation. For example, BS-Nets and DEHF obtain 38.2% and 56.4% OA on the IP dataset, respectively, both lower than FHBS-UML. Although existing cross-scene methods attempt to mitigate this by leveraging external data, they often suffer from domain shift issues, resulting in suboptimal transfer performance.

In contrast, our method achieves significant improvements by leveraging the meta-RL framework trained on the aforementioned source domains. On the challenging IP dataset, our method achieves an OA of 70.4%, significantly surpassing the second-best method by approximately 12.0%. On the SA and PU datasets, our method also obtains OAs of 89.7% and 82.9%, significantly exceeding the corresponding second-best methods by 6.7% and 7.4%, respectively. These consistent and substantial margins across different target scenes indicate that the performance gains of FHBS-UML are significant and stable, rather than resulting from random fluctuations. The superiority of FHBS-UML is also reflected in the classification maps. As visualized in Figs. 4–6, our approach produces results with higher regional consistency and significantly fewer misclassified pixels than other competing methods.

D. Parameter Sensitivity Analysis

1) Sensitivity Analysis on the Number of Shots: To further investigate the robustness of the proposed method under varying degrees of label scarcity, we conducted comprehensive experiments on the IP dataset by varying the number of labeled samples (shots) across the set $\{1, 3, 5, 10, 20, 50\}$ per class. The quantitative results are visualized in Fig. 7.

As illustrated in the figure, the proposed method consistently outperforms all competing baselines across all tested few-shot settings. Specifically, compared with the second-best performing methods in each setting, our method achieves substantial accuracy improvements of 13.22%, 11.93%, 12.03%, 6.57%,

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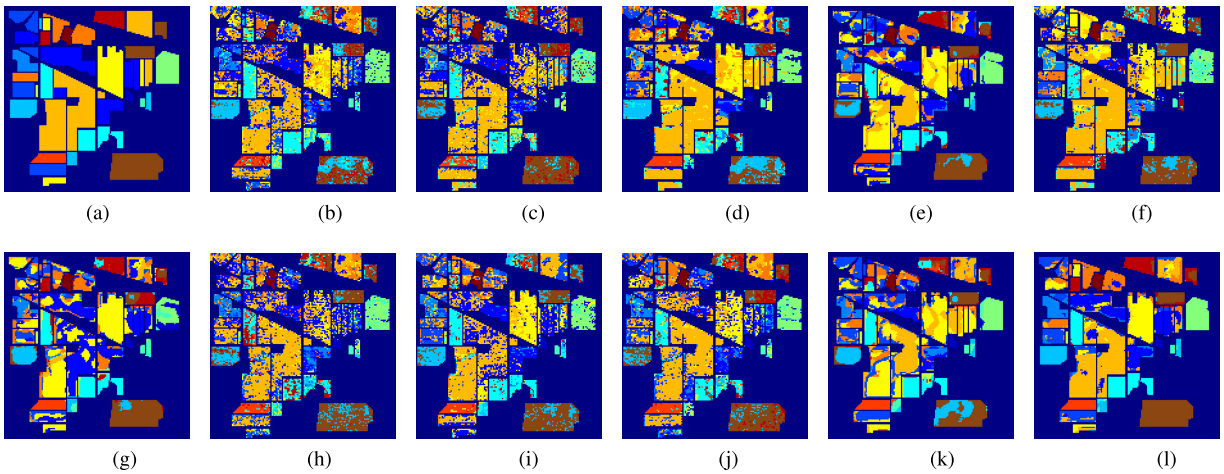


Fig. 4. Classification maps of different band selection methods on the IP dataset under the five-shot setting. (a) Ground truth. (b) MRMR. (c) BS-Nets. (d) DDCNN. (e) BHCNN-CFL. (f) CAE-UBS. (g) RLSBS. (h) SRL-SOA. (i) GWOGA. (j) EMAC-BS. (k) CSSRnet. (l) Ours.

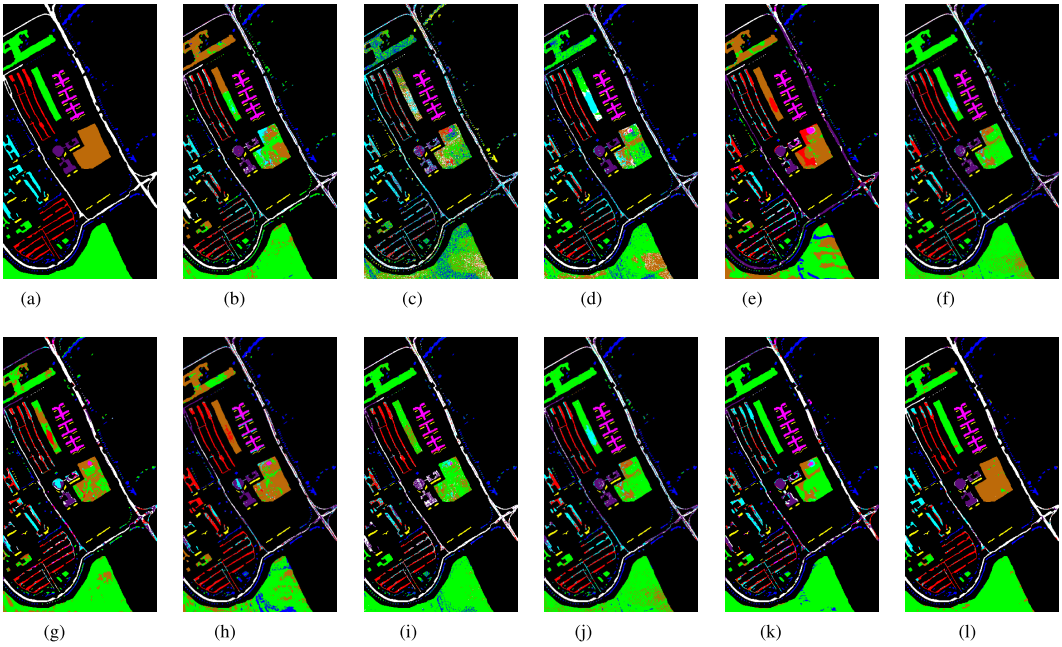


Fig. 5. Classification maps of different band selection methods on the PU dataset under the five-shot setting. (a) Ground truth. (b) MRMR. (c) BS-Nets. (d) DDCNN. (e) BHCNN-CFL. (f) CAE-UBS. (g) RLSBS. (h) SRL-SOA. (i) GWOGA. (j) EMAC-BS. (k) CSSRnet. (l) Ours.

4.34%, and 3.13% under the 1-, 3-, 5-, 10-, 20-, and 50-shot settings, respectively. In the extremely low-data regime, particularly the one-shot scenario, our approach maintains a distinct performance margin.

Analyzing the performance curves further reveals baseline limitations. Methods like CSSRnet plateau because their unsupervised reconstruction objectives overshadow the classification task as labeled samples increase. Conversely, FHBS-UML maintains a steeper accuracy curve, even at the one-shot extreme. This is because our variance-sensitive evaluator directly optimizes the few-shot discriminative quality, guiding the agent to select highly separable bands. Conversely, data-intensive deep learning methods such as BS-Nets and DDCNN suffer from severe overfitting and perform poorly in extremely low-data regimes, such as the one-shot setting. Nevertheless, their performance improves substantially as the sample size reaches 50, which unlocks their powerful feature extraction capabilities. Our method effectively bridges this gap by combining the rapid adaptation of meta-learning with the scalability of deep neural networks. It demonstrates a significant performance advantage in the one-shot setting and reaches convergence at the 20-shot setting. Even as the number of labeled samples increases to 50, our method maintains its lead. This indicates that our algorithm extracts sufficient discriminative information from as few as 20 samples to maximize accuracy, thereby significantly reducing the dependency on large-scale annotations.

2) Sensitivity Analysis on the Number of Selected Bands: To examine the effect of the number of selected bands, we conduct a sensitivity analysis under different band budgets, as shown in Fig. 8. The results show that classification performance increases rapidly when the band budget

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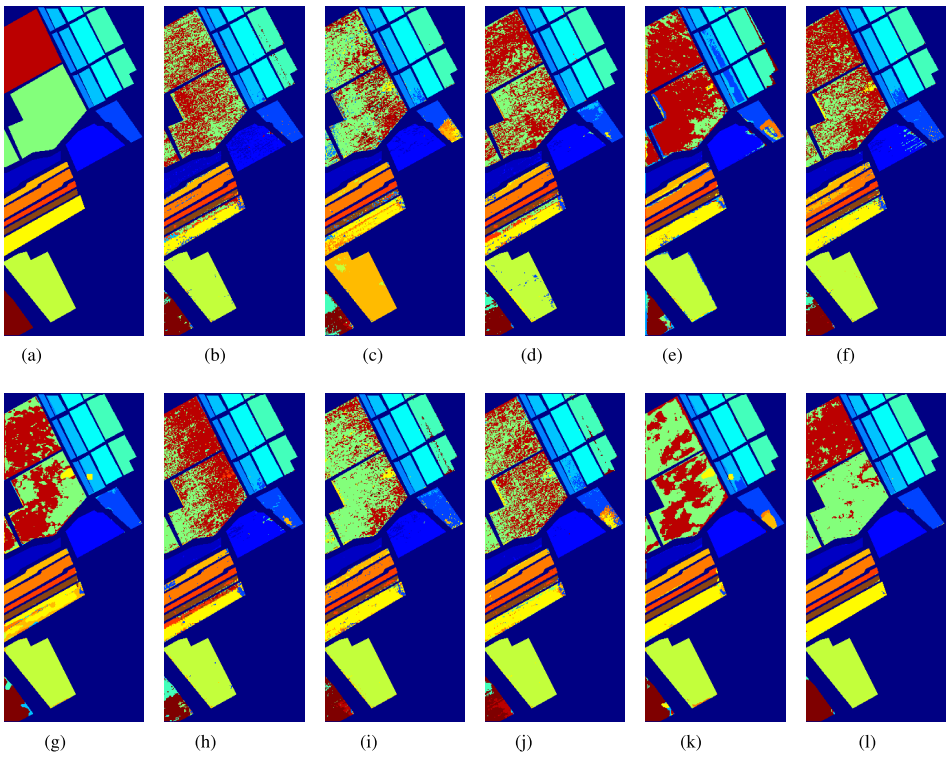


Fig. 6. Classification maps of different band selection methods on the SA dataset under the five-shot setting. (a) Ground truth. (b) MRMR. (c) BS-Nets. (d) DDCNN. (e) BHCNN-CFL. (f) CAE-UBS. (g) RLSBS. (h) SRL-SOA. (i) GWOGA. (j) EMAC-BS. (k) CSSRnet. (l) Ours.

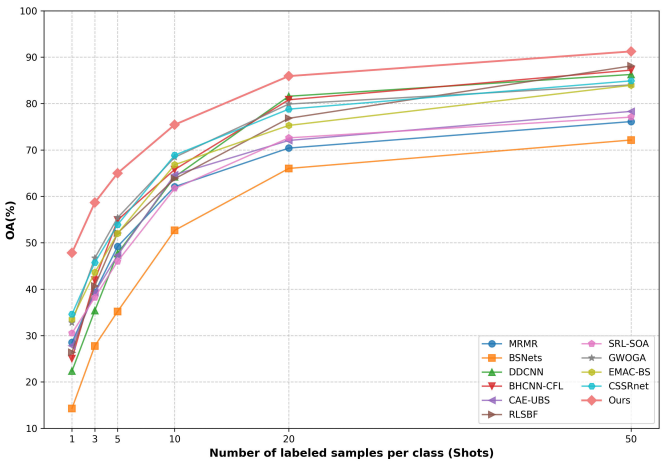


Fig. 7. Performance comparison of different methods on the IP dataset under different shots.

grows from a small value to the default setting, and then gradually saturates with more selected bands. When K is increased from the default setting to the full-band setting, the OA changes only marginally on IP, SA, and PU. These results indicate that the default compact band subsets have already captured most discriminative spectral information. Therefore, we set 15 bands for SA and 20 bands for IP and PU to balance classification accuracy and spectral compactness.

E. Ablation Study

To verify the contribution of the key modules in FHBS-UML, we conduct a series of ablation experiments. All experiments follow the cross-scene five-shot band selection protocol described in Section IV and are evaluated on the three target datasets: IP, SA, and PU.

1) Overall Ablation Analysis: We conduct an overall ablation study to evaluate the contribution of the major components in FHBS-UML, including HSA, the variance-sensitive evaluator, uncertainty-aware weighting, and the meta-learning mechanism. All variants follow the same source-target split, five-shot protocol, band-number setting, and random evaluation protocol as the full model.

Table IV summarizes the overall ablation results. "w/o HSA" replaces the proposed HSA with a standard 3-D-CNN encoder without alignment loss. "w/o variance-sensitive evaluator" uses the Euclidean distance instead of the proposed variance-sensitive metric. "w/o uncertainty weighting" adopts equal task weights, and "w/o meta-learning" replaces the meta-learning updates with standard joint training.

These ablation results provide a deeper explanation for the performance gains of the full FHBS-UML model. As shown in Table IV, removing any major component consistently degrades the performance, indicating that the effectiveness of FHBS-UML comes from the joint effects of heterogeneous representation alignment, robust few-shot evaluation, uncertainty-guided optimization, and meta-learning-based rapid adaptation. Removing HSA causes the largest performance drop, indicating that heterogeneous representation alignment is crucial for reducing cross-scene spectral discrepancies. Replacing the variance-sensitive evaluator weakens the model because the proposed evaluator provides more stable and reliable similarity estimation under limited support samples. Removing uncertainty-aware weighting reduces the

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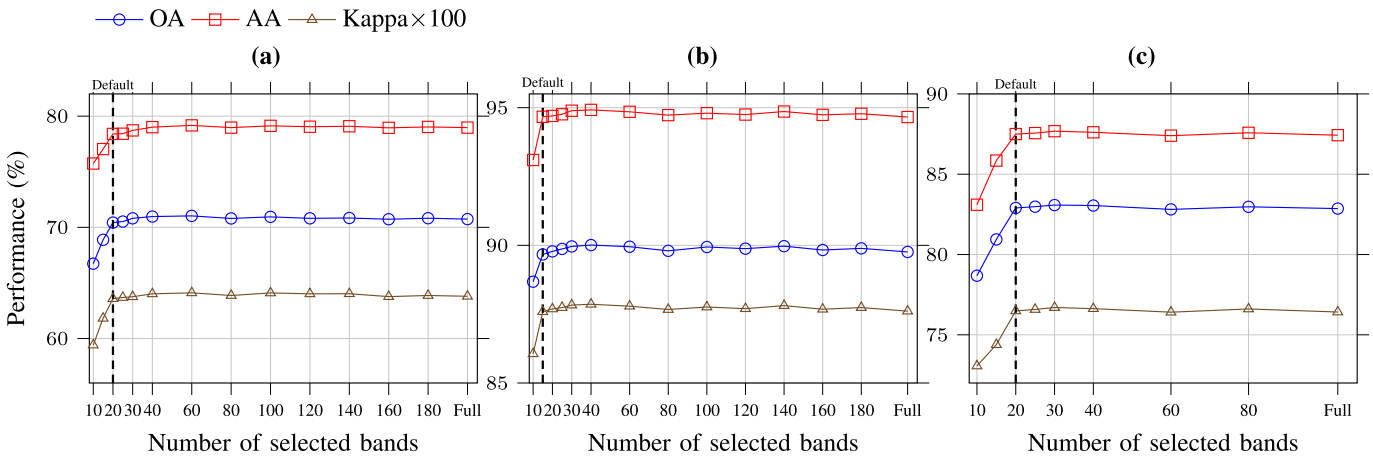


Fig. 8. Parameter sensitivity analysis of the number of selected bands on classification performance. (a) IP. (b) SA. (c) PU.

TABLE IV
COMPREHENSIVE ABLATION STUDY OF THE MAJOR COMPONENTS UNDER THE CROSS-SCENE FIVE-SHOT SETTING

Method	IP			SA			PU		
	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100
w/o HSA	60.37	69.29	54.63	82.18	87.44	79.51	72.04	77.26	63.53
w/o Variance-Sensitive Evaluator	66.88	75.12	59.95	86.95	91.92	84.74	79.06	83.89	71.83
w/o Uncertainty Weighting	66.18	75.42	58.84	86.74	90.15	83.46	78.86	82.41	69.13
w/o Meta-learning	64.35	73.46	56.91	85.74	89.28	82.07	76.58	80.42	67.41
FHBS-UML (Full)	**70.43**	**78.40**	**63.59**	**89.67**	**94.67**	**87.59**	**82.90**	**87.50**	**76.49**

TABLE V
ABLATION STUDY OF THE HETEROGENEOUS HSA ON IP, SA, AND PU DATASETS

Data	Encoder	Loss Function	OA (%)	AA (%)	Kappa × 100
IP	3D-CNN	w/o Alignment Loss	60.37	69.29	54.63
		CORAL Loss	63.15	72.04	57.88
		MMD Loss	64.42	72.93	59.06
	HSE	w/o Alignment Loss	66.08	73.71	59.37
		CORAL Loss	68.58	76.21	61.17
		MMD Loss (Ours)	**70.43**	**78.40**	**63.59**
SA	3D-CNN	w/o Alignment Loss	82.18	87.44	79.51
		CORAL Loss	84.74	89.52	82.03
		MMD Loss	86.17	90.36	83.76
	HSE	w/o Alignment Loss	86.45	90.53	83.93
		CORAL Loss	88.33	92.86	85.94
		MMD Loss (Ours)	**89.67**	**94.67**	**87.59**
PU	3D-CNN	w/o Alignment Loss	72.04	77.26	63.53
		CORAL Loss	74.87	79.92	66.55
		MMD Loss	76.65	81.23	68.54
	HSE	w/o Alignment Loss	78.56	82.81	70.74
		CORAL Loss	80.92	85.47	73.79
		MMD Loss (Ours)	**82.90**	**87.50**	**76.49**

performance, showing that adaptively suppressing uncertain source tasks helps stabilize meta-training. Disabling meta-learning also leads to inferior results, confirming that the proposed meta-learning mechanism is essential for fast adaptation to unseen few-shot target scenes.

2) Effectiveness of the HSA: After evaluating the overall contribution of HSA, we further analyze its internal design from two aspects: the feature encoder architecture and the feature alignment strategy. The quantitative comparisons on the IP, SA, and PU datasets are summarized in Table V.

We compare the proposed HSE against a standard 3-D-CNN baseline. As shown in Table V, the HSE-based method consistently outperforms the 3-D-CNN counterpart across all datasets. Specifically, on the IP dataset with MMD loss, the HSE achieves an average OA of 70.43%, surpassing the 3-D-CNN (64.42%) by a significant margin of 6.01%. Similar trends are observed on SA and PU. These results demonstrate that the attention mechanism in HSE captures spectral dependencies more effectively than standard convolutions, yielding robust representations for heterogeneous data.

To assess the alignment mechanism, we further examine the effect of feature alignment in Table V. The results show that introducing MMD loss consistently improves the performance across all three datasets. Compared with the model without alignment, MMD loss improves the OA by 4.35%, 3.22%, and 4.34% on IP, SA, and PU, respectively. This confirms that reducing distributional discrepancies is beneficial for few-shot knowledge transfer across heterogeneous HSI datasets.

3) Effectiveness of the Stability-Enhanced Heterogeneous HSI Evaluator: To validate the stability-enhanced heterogeneous HSI evaluator, we investigate the impact of metric space selection in Table VI. Unlike the standard Euclidean distance, which fails to capture the anisotropic distributions of heterogeneous data due to its rigid spherical assumption, our proposed Variance-Sensitive measure explicitly integrates second-order statistical constraints into the metric learning process. By dynamically rectifying the distance calculation with adaptive covariance estimates, our method robustly handles unstable

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TABLE VI

ABLATION STUDY OF DIFFERENT DISTANCE METRICS ON CLASSIFICATION PERFORMANCE

Data	Condition	OA (%)	AA (%)	Kappa × 100
IP	Euclidean distance	66.88	75.12	59.95
	Manhattan distance	68.15	76.47	61.53
	Variance-sensitive	**70.43**	**78.40**	**63.59**
SA	Euclidean distance	86.95	91.92	84.74
	Manhattan distance	88.14	93.18	86.05
	Variance-sensitive	**89.67**	**94.67**	**87.59**
PU	Euclidean distance	79.06	83.89	71.83
	Manhattan distance	80.82	85.45	73.91
	Variance-sensitive	**82.90**	**87.50**	**76.49**

TABLE VII

ABLATION STUDY OF MULTIOBJECTIVE TASK WEIGHTING STRATEGIES (λ) ON IP, SA, AND PU DATASETS

Dataset	Weighting Strategy	OA (%)	AA (%)	Kappa×100
IP	Equal Weights ($\lambda_i = 1/K$)	66.18	75.42	58.84
	Adaptive Weights (Ours)	**70.43**	**78.40**	**63.59**
SA	Equal Weights ($\lambda_i = 1/K$)	86.74	90.15	83.46
	Adaptive Weights (Ours)	**89.67**	**94.67**	**87.59**
PU	Equal Weights ($\lambda_i = 1/K$)	78.86	82.41	69.13
	Adaptive Weights (Ours)	**82.90**	**87.50**	**76.49**

intraclass variations, achieving significant OA improvements of 3.55%, 2.72%, and 3.84% over the Euclidean baseline on the IP, SA, and PU datasets, respectively.

4) Effectiveness of Uncertainty-Aware Multiobjective Meta-Learning Optimization: Finally, we investigate the impact of the task weighting strategy within the meta-learning optimization framework. Table VII presents the performance comparison between the standard equal weights strategy ($\lambda_i = 1/K$) and our proposed adaptive weights strategy.

Conventional meta-learning approaches, such as MAML, typically assign uniform importance to all tasks during the gradient update. However, this assumption overlooks the inherent variability in task difficulty across heterogeneous source datasets, potentially leading to suboptimal generalization. In contrast, our proposed multiobjective optimization leverages homoscedastic uncertainty to dynamically learn the task-specific weights (λ_i) during training. This mechanism automatically balances the contribution of multiple loss functions without manual hyperparameter tuning. As evidenced in Table VII, the adaptive weights strategy consistently outperforms the equal weights baseline. These results confirm that accounting for task uncertainty effectively prevents the agent from being dominated by simple tasks or misled by noisy ones, thereby enhancing the generalized agent learning.

5) Effectiveness of the Meta-Learning Mechanism: We further analyze the contribution of the meta-learning mechanism using the results in Table IV. Compared with the full model, removing meta-learning consistently degrades the performance on all three datasets, which suggests that meta-learning is beneficial for learning transferable band selection policies that can better adapt to unseen target sensors under the few-shot setting.

TABLE VIII

COMPUTATIONAL COST OF DIFFERENT BAND SELECTION METHODS ON THE IP DATASET UNDER THE FIVE-SHOT SETTING

Method	Target Training Time(s)	Inference Time(s)
MRMR	7.1 ± 0.4	0.5 ± 0.1
BS-Nets	678.2 ± 15.1	1.5 ± 0.1
DDCNN	486.6 ± 10.8	1.3 ± 0.1
BHCNN-CFL	537.4 ± 7.9	1.4 ± 0.1
CAE-UBS	462.7 ± 12.6	1.5 ± 0.1
RLSBS	873.3 ± 18.4	1.1 ± 0.1
SRL-SOA	249.8 ± 10.1	0.7 ± 0.1
GWOGA	346.1 ± 10.8	0.7 ± 0.1
DEHF	643.8 ± 15.4	1.6 ± 0.1
EMAC-BS	279.5 ± 9.2	0.8 ± 0.1
CSSRnet	742.5 ± 13.6	1.1 ± 0.1
Ours	371.6 ± 10.3	0.9 ± 0.1

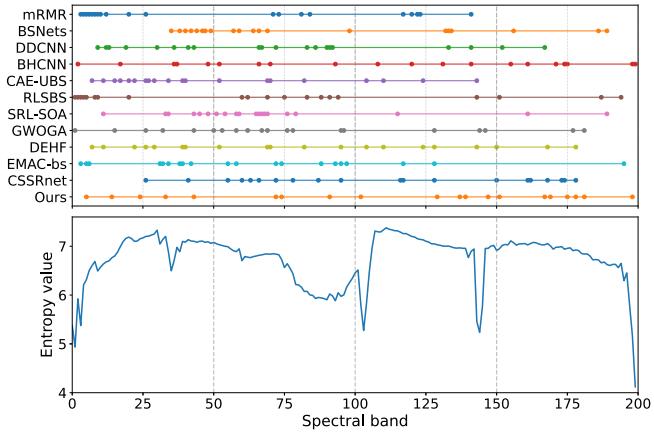


Fig. 9. Visualization of selected bands on the IP dataset. The upper panel shows the selected-band distributions of different methods, and the lower panel shows the spectral entropy curve.

F. Computational Efficiency Analysis

Table VIII reports the running time of different band selection methods on the IP target dataset under the same few-shot setting, where target training time refers to target-domain training, adaptation, or band selection before inference, and inference time denotes classification using the selected bands. All results are reported as mean ± standard deviation over ten independent runs. Although classical and heuristic methods are generally faster, FHBS-UML achieves a moderate training cost and comparable inference time among deep learning-based methods, while delivering superior classification performance. These results indicate a favorable tradeoff between computational efficiency and accuracy under the few-shot heterogeneous setting.

G. Selected Band Analysis

Fig. 9 visualizes the selected-band distributions of different methods on the IP dataset along the spectral dimension, together with the corresponding spectral entropy curve. All band indices use zero-based indexing and correspond to the valid bands after removing water-absorption bands.

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TABLE IX
PERFORMANCE UNDER BALANCED FIVE-SHOT AND PROPORTIONAL IMBALANCED SAMPLING ON IP

Method	Balanced 5-Shot			Proportional Sampling		
	OA	AA	K×100	OA	AA	K×100
MRMR	52.2±2.1	59.4±1.7	46.4±2.1	55.1±2.6	51.8±3.4	50.9±2.5
BS-Nets	38.2±2.6	41.4±1.4	37.0±2.6	42.9±2.5	36.2±3.6	37.8±3.1
DDCNN	50.7±2.5	61.6±3.2	44.7±2.0	56.8±3.0	53.2±3.8	50.2±2.9
BHCNN-CFL	57.9±3.3	73.0±2.5	53.7±2.9	63.1±3.1	61.2±3.9	59.4±3.4
CAE-UBS	50.3±2.7	61.9±3.1	44.3±2.3	56.9±3.2	51.6±3.7	51.9±3.1
RLSBS	55.1±2.2	58.9±3.5	45.6±2.0	62.2±3.1	56.4±4.2	57.1±3.2
SRL-SOA	49.0±2.8	52.7±2.8	42.6±2.9	55.7±3.3	48.5±3.5	51.4±3.3
GWOGA	58.4±3.3	64.4±3.3	52.4±2.9	63.6±3.7	57.0±3.9	58.5±3.6
DEHF	56.4±3.1	63.8±2.9	51.9±3.4	62.7±3.5	56.4±3.6	57.8±3.8
EMAC-BS	55.0±3.7	63.4±3.4	48.4±3.4	61.2±4.0	56.3±4.2	54.8±3.9
CSSRnet	56.9±3.2	67.9±3.5	52.9±2.9	63.4±3.6	58.8±4.0	56.7±3.6
Ours	**70.4±1.6**	**78.4±1.4**	**63.6±1.2**	**76.1±1.9**	**72.4±2.3**	**69.3±1.6**

The compared methods exhibit distinct selection patterns. Several baselines either concentrate on local spectral regions or show irregular distributions across the spectrum. In contrast, FHBS-UML selects bands from regions with low, intermediate, and high spectral indices, leading to more balanced spectral coverage. The selected bands also largely coincide with high-entropy regions, indicating that FHBS-UML preserves complementary spectral information while avoiding over-concentration within a narrow spectral interval.

The final band subset is obtained by ranking band occurrence frequencies over ten independent runs. On IP, nine FHBS-UML-selected bands appear in all ten runs, and the remaining bands appear at least six times, confirming the stability of the selected subset.

H. Evaluation Under Class-Imbalanced Few-Shot Sampling

To further evaluate the robustness of FHBS-UML under realistic class-imbalanced conditions, we conduct an additional experiment on the IP dataset by comparing two support-set sampling strategies. The first is the standard balanced five-shot setting described in Section IV-C. The second is a proportional imbalanced sampling strategy, where the total number of support samples is fixed at 80 and allocated to each class according to its proportion in the full dataset, with at least one sample assigned to each class. Specifically, the per-class sample counts for classes 1–16 are 1, 10, 6, 2, 4, 6, 1, 4, 1, 7, 18, 5, 2, 9, 3, and 1, respectively. The mean and standard deviation over ten independent runs under both strategies are reported in Table IX. The results show that the adaptive class-specific covariance calibration ($\mathbf{Q}_k$) helps maintain discriminative performance under class-imbalanced few-shot conditions.

I. SVM Evaluation of Selected Band Subsets

To isolate the influence of the final classifier, we further conduct a classifier-independent evaluation. After each method obtains its selected band subset under the cross-scene five-shot setting, all selected subsets are re-evaluated using the same SVM classifier under identical data splits, preprocessing, and hyperparameter selection. The proposed evaluator and method-specific final classifiers are excluded.

TABLE X
SVM EVALUATION OF SELECTED BAND SUBSETS UNDER THE CROSS-SCENE FIVE-SHOT SETTING

Method	Indian Pines	Salinas	Pavia Univ.
MRMR	52.4 ± 3.1	78.5 ± 2.2	66.4 ± 3.2
BS-Nets	38.1 ± 3.5	64.1 ± 3.2	48.7 ± 3.4
DDCNN	48.6 ± 2.5	78.5 ± 3.3	64.8 ± 3.1
BHCNN-CFL	55.4 ± 3.3	69.8 ± 3.1	56.5 ± 2.9
CAE-UBS	50.2 ± 2.8	82.7 ± 2.6	75.2 ± 3.2
RLSBS	53.5 ± 2.6	80.8 ± 3.3	71.6 ± 2.3
SRL-SOA	49.2 ± 2.4	80.6 ± 2.6	67.4 ± 3.5
GWOGA	58.1 ± 3.2	77.9 ± 2.4	71.2 ± 3.8
DEHF	56.4 ± 3.1	78.9 ± 3.2	74.1 ± 2.7
EMAC-BS	53.2 ± 3.2	75.1 ± 2.8	70.2 ± 2.2
CSSRnet	54.3 ± 2.7	76.9 ± 3.0	71.3 ± 2.5
Ours	**66.8 ± 2.4**	**86.9 ± 2.1**	**79.6 ± 2.3**

TABLE XI
CROSS-SCENE OA COMPARISON ACROSS ADDITIONAL TARGET DATASETS UNDER THE FIVE-SHOT SETTING

Method	Botswana	KSC	Houston	WDC
MRMR	61.2	58.6	63.8	64.5
BS-Nets	50.8	52.1	52.6	56.9
DDCNN	58.9	60.4	64.7	63.8
BHCNN-CFL	66.1	62.8	69.1	67.5
CAE-UBS	66.8	67.4	73.8	74.6
RLSBS	70.8	65.7	75.8	78.6
SRL-SOA	59.6	60.7	67.7	66.1
GWOGA	67.9	64.3	71.6	70.8
DEHF	70.2	66.9	74.6	78.3
EMAC-BS	67.5	64.9	70.2	74.3
CSSRnet	69.8	67.6	73.6	78.4
Ours	**77.2**	**74.8**	**82.7**	**84.1**

As shown in Table X, FHBS-UML achieves the best OA on all three target datasets under the standardized SVM protocol, indicating that its selected bands remain highly discriminative even without the customized evaluator.

J. Cross-Scene Evaluation on Additional Target Datasets

To further evaluate the effectiveness of FHBS-UML on other unseen target datasets, we conduct additional cross-scene experiments on Botswana, KSC, University of Houston, and WDC. Different from the main cross-scene setting where IP, SA, and PU are used as target domains, this experiment uses IP, SA, and PU as source domains for meta-training, while Botswana, KSC, Houston, and WDC are held out as target domains for evaluation. Under this setting, the meta-training process is restarted from scratch, with all model parameters and optimizer states reinitialized. No checkpoint or learned parameters obtained from the main source-target setting are reused. The same five-shot evaluation protocol is adopted for all target domains.

The quantitative results are summarized in Table XI. FHBS-UML achieves the best OA on all four additional target datasets, obtaining 77.2%, 74.8%, 82.7%, and 84.1% on Botswana, KSC, Houston, and WDC, respectively. Compared with the strongest baseline on each dataset, the corresponding OA gains are 6.4%, 7.2%, 6.9%, and 5.5%. The consistent

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improvements across both natural and urban scenes show that FHBS-UML learns transferable band selection policies rather than overfitting to a specific source-target split.

V. Conclusion

In this article, we proposed a novel few-shot heterogeneous hyperspectral band selection framework, termed FHBS-UML, from a meta-RL perspective. To overcome the challenges of domain shifts and label scarcity, our method integrates three key innovative components. First, to unify feature representations across diverse sensors, we designed an HSA utilizing a local–nonlocal encoder and MMD constraints. Second, to address the instability of intraclass variance in heterogeneous data, we introduced a stability-enhanced heterogeneous HSI evaluator equipped with a variance-sensitive metric to rectify classification confidence. Third, to enhance generalization across tasks with varying difficulty, we developed an uncertainty-aware multiobjective meta-learning optimization strategy that leverages homoscedastic uncertainty to adaptively learn task-specific weights. Comprehensive experiments on the IP, SA, and PU datasets demonstrate that FHBS-UML consistently outperforms state-of-the-art methods across a wide range of few-shot settings, validating its superior adaptability and robustness in heterogeneous hyperspectral applications. Beyond the current evaluation, the application of FHBS-UML to broader real-world scenarios remains worth further exploration, particularly for more diverse sensor characteristics and complex cross-domain variations. Future work will investigate more efficient meta-training and adaptation strategies, and further extend the framework to heterogeneous scenarios involving larger sensor gaps and more diverse land-cover categories.

References

[1] G.Hughes, "On the mean accuracy of statistical pattern recognizers," *IEEE Trans. Inf. Theory*, vol. IT-14, no. 1, pp. 55–63, Jan.1968.

[2] W.Sunand Q.Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun.2019.

[3] B.Han et al., "Hyperspectral band selection with unique pixel extraction and adaptive neighbor clustering," *Remote Sens.*, vol. 17, no. 2, p. 315, Jan.2025.

[4] L.Gao, X.Xue, and H.Zheng, "Sparse hyperspectral band selection based on expectation maximization," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 1, pp. 1265–1278, Jan.2026.

[5] L.Zhang, Y.Wei, J.Liu, J.Wu, and D.An, "A hyperspectral band selection method based on sparse band attention network for maize seed variety identification," *Expert Syst. Appl.*, vol. 238, Mar.2024, Art. no. 122273.

[6] K.Bi et al., "Open-set domain adaptation for hyperspectral image classification based on weighted generative adversarial networks and dynamic thresholding," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5507717.

[7] F.Zhuang et al., "A comprehensive survey on transfer learning," *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, Jan. 2021.

[8] Y.Zhang, W.Li, R.Tao, J.Peng, Q.Du, and Z.Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9646–9660, Nov.2021.

[9] W.Zhang, Z.Wang, J.Ren, R.Shang, and L.Jiao, "Few-shot hyperspectral image classification based on prototype mask contrast and cross-stage feature refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5520318.

[10] J.Feng, G.Bai, D.Li, X.Zhang, R.Shang, and L.Jiao, "MR-selection: A meta-reinforcement learning approach for zero-shot hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500320.

[11] X.He, Y.Chen, and P.Ghamisi, "Heterogeneous transfer learning for hyperspectral image classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3246–3263, May2020.

[12] Y.Peng, Y.Liu, B.Tu, and Y.Zhang, "Convolutional transformer-based few-shot learning for cross-domain hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1335–1349, 2023.

[13] L.Ma, Y.Yang, G.Wang, Z.Li, W.Sun, and Q.Du, "Open-set domain adaptation for hyperspectral image classification based on reconstruction discrepancy and feature alignment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 24829–24842, 2025.

[14] M.Ye, Y.Xu, H.Lu, K.Yan, and Y.Qian, "Cross-scene feature selection for hyperspectral images based on cross-domain information gain," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 4764–4767.

[15] C.Zhang, M.Ye, L.Lei, and Y.Qian, "Feature selection for cross-scene hyperspectral image classification using cross-domain I-ReliefF," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5932–5949, 2021.

[16] J.Wang, M.Ye, F.Xiong, and Y.Qian, "Cross-scene hyperspectral feature selection via hybrid whale optimization algorithm with simulated annealing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2473–2483, 2021.

[17] H.Peng, F.Long, and C.Ding, "Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 8, pp. 1226–1238, Aug.2005.

[18] J.Feng et al., "Convolutional neural network based on bandwise-independent convolution and hard thresholding for hyperspectral band selection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4414–4428, Sep.2021.

[19] Y.Cai, X.Liu, and Z.Cai, "BS-nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Mar.2020.

[20] H.Zhang, Y.Li, Y.Jiang, P.Wang, Q.Shen, and C.Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug.2019.

[21] J.Feng et al., "Deep reinforcement learning for semisupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2022.

[22] J.Aravinth, B. M. P.Reddy, S.Deenadayalan, S.Rajesh, and U.Bharadwaj, "Hyperspectral band selection using SRL-SOA algorithm," in *Proc. 7th Int. Conf. Energy, Power Environ. (ICEPE)*, May 2025, pp. 1–6.

[23] Y.Shang, M.Zheng, J.Li, and X.Zheng, "An effective feature selection approach based on hybrid grey wolf optimizer and genetic algorithm for hyperspectral image," *Sci. Rep.*, vol. 15, no. 1, p. 1968, 2025.

[24] J.Zhang, J.Wang, D.Li, J.Zhang, X.Sun, and X.Shang, "Hyperspectral band selection with dynamic graph enhancement and hierarchical feature fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5508816.

[25] C.Tang et al., "Spatial and spectral structure preserved self-representation for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531413.

[26] H.Sun, J.Ren, H.Zhao, P.Yuen, and J.Tschannerl, "Novel Gumbel–Softmax trick enabled concrete autoencoder with entropy constraints for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506413.

[27] W.Sun, K.He, G.Yang, J.Peng, K.Ren, and J.Li, "A cross-scene self-representative network for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509212.

[28] S.Behera, P. P.Sarangi, and B. S. P.Mishra, "An enhanced Jaya algorithm with mutation and diversity-preserving strategies for hyperspectral band selection," *F1000Research*, vol. 14, p. 999, Jan.2025.

[29] Y.Zhang, L.Li, Q.Lin, J.Li, Z.Ming, and V. C.Leung, "Evolutionary multitasking with adaptive cross-dataset knowledge transfer for band selection of hyperspectral images," *IEEE Trans. Emerg. Topics Comput. Intell.*, early access, Aug. 2024, doi: 10.1109/TETCI.2024.3442859.

[30] T.Hospedales, A.Antoniou, P.Micaelli, and A.Storkey, "Meta-learning in neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5149–5169, Sep.2022.

[31] C.Finn, P.Abbeel, and S.Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 1126–1135.

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[32] Y.Zhang, M.Zhang, W.Li, S.Wang, and R.Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.
[33] J.Feng et al., "Attention multibranch convolutional neural network for hyperspectral image classification based on adaptive region search," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5054–5070, Jun.2021.
[34] F.Zhou, R.Hang, Q.Liu, and X.Yuan, "Hyperspectral image classification using spectral–spatial LSTMs," *Neurocomputing*, vol. 328, pp. 39–47, Feb.2019.
[35] A.Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
[36] S.Li et al., "Enhancing the locality and breaking the memory bottleneck of transformer on time series forecasting," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–13.
[37] A.Gretton, K. M.Borgwardt, M. J.Rasch, B.Schölkopf, and A.Smola, "A kernel two-sample test," *J. Mach. Learn. Res.*, vol. 13, no. 1, pp. 723–773, 2012.
[38] J.Wangand Y.Zhai, "Prototypical Siamese networks for few-shot learning," in *Proc. IEEE 10th Int. Conf. Electron. Inf. Emergency Commun. (ICEIEC)*, Jul. 2020, pp. 178–181.
[39] R.Cipolla, Y.Gal, and A.Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7482–7491.
[40] Y.Zhan, D.Hu, H.Xing, and X.Yu, "Hyperspectral band selection based on deep convolutional neural network and distance density," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2365–2369, Dec.2017.



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